

Instructor: Date:

Learner: Regional Branch:

Program: Center:

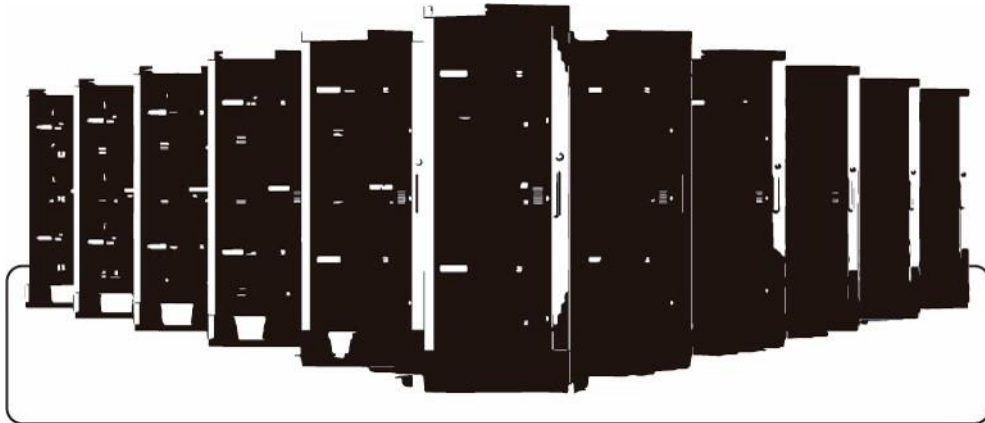
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Let's Start!

This is the data warehouse architecture.



Learning Outcome:

Designs the data structure based on the database structures and its characteristics.

Understand the concept of data warehouse, its components and characteristics.

Can Do Statements:

- I can talk about single layer architecture.
- I can describe second layer architecture.
- I can give information about third layer architecture.

Ask your partner:

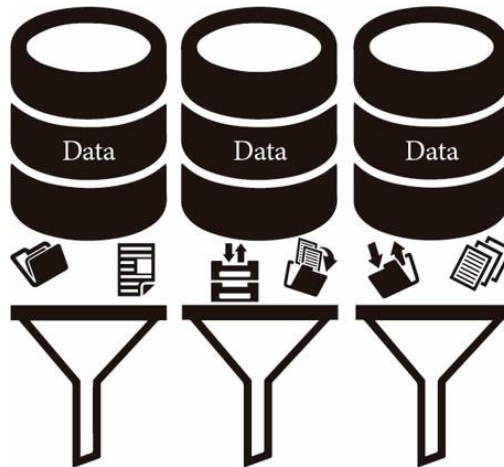
- What do you know about data warehouse architecture?
- What do you think "layer architecture" refers to?





Let's Explore!

Would you rather use the second layer or the third layer?



Instructions: Read the text below.

- Fred:** Hi, Monica. I'm changing the data warehouse architectures because the boss asked me so. However, I don't know how to do it; I don't know what layers I ought to use.
- Monica:** You might want to choose the layer that fulfills your needs and goals. Would you rather use a single, second, or third layer architecture?
- Fred:** Well, that's the issue. I don't have a clear picture of what they are. Would you mind instructing me?
- Monica:** No problem. Look, the first layer architecture aims at minimizing the amount of data available in the warehouse. Basically, it takes away any redundancy in information. There are weakness and strength. As you know, we might want to have analytical and transactional processes separated. In this case, this layer doesn't comply with this requirement. As a strength, you'd better use this layer if the amount of data to be analyzed is huge; of course, you need to make some arrangements for that to happen.
- Fred:** What about the second layer?
- Monica:** The second layer aims at the separation of the data warehouse. It separates physical sources from the data warehouse. As its name shows, the main idea of this layer is to have different data flow phases.
- Fred:** I guess that the third layer is more complex, am I right?
- Monica:** Yes, you are. The third layer materializes operational data grabbed after combining source data. The most important characteristic of this layer is that it creates a conventional data reference for the company. Also, it separates problems when extracting and integrating data.
- Fred:** Would you rather use a second or third layer architecture?
- Monica:** Well, I don't know. Would your boss rather prefer a single, second or third layer?

**Let's Practice!**

The characteristics of the second layer.

Instructions: Proofread the sentences. Correct the mistakes. Read the sentences aloud. Check your pronunciation with your instructor.

1. Could you rather use the second layer? Then remember that this layer has different phases to be followed.

2. The main characteristic of this layer is separate physical sources and data warehouse.

3. You would want to remember that it has 4 stages are: source layer, data staging, data warehouse layer and analysis.

4. You had better use this layer if you wants to have clearly organized and separated information.

**Let's Practice!**

The complexity and usefulness of the architecture layers

Instructions: Select the correct answer to complete the sentences.

1. Single-layer architecture main purpose is:
 - a. to minimize the amount of information available and it aims to avoid redundancy.
 - b. to separate physical sources and data warehouse.
 - c. that it creates a neutral data model for the company.
 - d. that it creates redundancy talking about operational data.
2. Second-layer architecture main characteristic is:
 - a. to minimize the amount of information available and it aims to avoid redundancy.
 - b. to separate physical sources and data warehouse.
 - c. that it creates a neutral data model for the company.
 - d. that it creates redundancy talking about operational data.





Let's Practice!

3. Third-layer architecture main feature is:
 - a. to minimize the amount of information available and it aims to avoid redundancy.
 - b. to separate physical sources and data warehouse.
 - c. that it creates a neutral data model for the company.
 - d. that it creates redundancy talking about operational data.
4. One third-layer architecture disadvantage is:
 - a. to minimize the amount of information available and it aims to avoid redundancy.
 - b. to separate physical sources and data warehouse.
 - c. that it creates a neutral data model for the company.
 - d. that it creates redundancy talking about operational data.
5. It is the main feature of third-layer architecture:
 - a. It creates a neutral data model for the company.
 - b. It minimizes the amount of information available.
 - c. It aims to avoid redundancy.
 - d. to separate physical sources and data warehouse.



Extension Activity

The data warehouse layers in a real life context.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- a) Look for a company that uses data warehouse layers.
- b) Look for which layer and how the company uses it.
- c) Analyze the pros and cons of using that layer in that company.
- d) Make a chart comparing the pros and cons.
- e) Present your work to the class.

Self-Assessment

- | | Yes | No |
|--|-----------------------|-----------------------|
| 1. I find a company that uses data warehouse layers. | ☐ | ☐ |
| 2. I look for the way they use the layers. | ☐ | ☐ |
| 3. I analyze the pros and cons of the layer in that company. | ☐ | ☐ |
| 4. I make a chart comparing my findings. | ☐ | ☐ |
| 5. I share my work with my classmates. | ☐ | ☐ |



Target Words

- **weakness** (noun): lack of strength
e.g. There is a weakness and a strength; as you know, we might want to have analytical and transactional processes separated, in this case this layer doesn't comply with this requirement.
- **grab** (verb): to obtain something
e.g. The third layer materializes operational data grabbed after combining source data.
- **redundancy** (noun): something that is repeated
e.g. Basically it takes away any redundancy of information.
- **need** (noun): a necessity
e.g. You ought to choose the layer that fulfills your needs and goals.
- **architecture** (noun): the structure of something
e.g. Hi Monica, I'm changing the data warehouse architectures because the boss asked me to.
- **available** (adjective): that is ready for use
e.g. Look, the first layer architecture aims to minimize the amount of data available in the warehouse.



Learning Points

Would Rather for What is Preferable: Interrogative.

You can use the WOULD RATHER to talk about something that you prefer.

When you use WOULD RATHER in questions you need to keep in mind that you are giving some options to be chosen.

To create an interrogative statement using WOULD RATHER you need to use the following

Structure:

Would + Pronoun + **Rather** + Verb + Complement + ?

Example: **Would** you **rather** use a single or a second layer architecture?

Examples

- *Would you rather use a single, second or third layer architecture?*
- *Well, I don't know. Would your boss rather prefer a single, second or third layer?*





Learning Points

Data Warehouse Architecture Layers

According to the needs required by a company, it is possible to use different architecture layers for **data warehouse**.

Every **data warehouse** has the same properties: separation, scalability, extensibility, security and administerability. However, two different classifications are made according to the architecture layers used, these are enterprise oriented or department oriented.

The architecture layers to be used are: single-layer, second-layer and third-layer architecture. A single-layer architecture is not commonly used, yet its main purpose is to minimize the amount of information available and it aims to avoid redundancy.

The second-layer architecture has different phases to be followed, since its main characteristic is to separate physical sources and **data warehouse**. Its main stages are: source layer, data staging, **data warehouse** layer and analysis.

The third-layer architecture basically materializes findings after integrating data. Its main feature is that it creates a neutral data model for the company. One of its disadvantages is that it creates redundancy talking about operational data.

Examples

- *Look, the first layer architecture aims to minimize the amount of data available in the warehouse. Basically it takes away any redundancy of information. There is a weakness and a strength; as you know, we might want to have analytical and transactional processes separated, in this case this layer doesn't comply with this requirement. As a strength, you'd better use this layer just in case the amount of data to be analyzed is huge, of course you need to make some arrangements for that to happen.*
- *The second layer aims to the separation of data warehouse. It separates physical sources from data warehouse. As its name shows, the main idea of this layer is to have different data flow phases.*

